

Sentinel - Executive Summary

On-Premises Psychophysiological Triage Node

Prepared for incubation evaluation | 2026 | Companion: White Paper, Validation & Timing Report, Business Plan

1. The Product

Sentinel is a hardware-software infrastructure for continuous mental health support that connects three stakeholders in one real-time ecosystem: the patient, the psychologist, and the trusted contact. A patient portal captures biometric signals from wearable smart rings and AI-summarized journal entries. A clinician dashboard presents triage-ranked patients, AI-synthesized clinical notes, and a live crisis broadcast. A no-login trusted-contact page provides signed escalation links during crisis events. Every clinical loop - journal to insight, crisis to acknowledgement, task to graded proof - closes inside the platform.

Sentinel is designed to support mental health professionals, not replace clinical judgment.

2. The Problem

Therapy is confined to weekly sessions while mental health risk evolves continuously between them. Clinicians carry caseloads with heavy documentation burden, and physiological, textual, and clinical signals live in separate silos. Crisis support depends on appointment availability - and often on a single point of failure.

3. The Solution

- Fused signal pipeline: physiology (HR, HRV, stress, sleep, SpO2) + 28-label GoEmotions emotion classification + self-reported mood, correlated to surface cross-signal discrepancies.
- Deterministic crisis engine: staged escalation (patient siren, trusted-contact email, helpline), haltable by clinician acknowledgement at any point, with cooldown throttling.
- Privacy-first AI: primary inference runs fully offline via a local 7.2B-parameter therapy-tuned model (Ollama); cloud AI is opt-in.
- Zero-infrastructure cost: open-source stack (FastAPI + React PWA), SQLite/PostgreSQL, no per-seat or per-inference fees.
- Hardware-agnostic: pluggable RingSource SDK (BLE, vendor cloud, simulator); hardware M0 complete, OEM ring in procurement.

4. Validation Highlights

Claim	Measured result
Automated test suite	98/98 tests passed, 12 modules, 0 skips
Discrepancy accuracy	96.0% (Prec 91%, Rec 100%), 0.1 ms per evaluation
Crisis engine concurrency	102-109 ms at 1-25 concurrent, 0 dropped threads
Halt protocol	Stages fire correctly at 15/45/65 s ack points
Local AI summarization	1.4-1.7 s warm, ~5.1 s cold (Mistral, commodity hardware)

REST API latency	7-41 ms on live deployment
Full AI triage pipeline	~4.7 s end-to-end incl. one local LLM call
Security primitives	PBKDF2 4-156 ms; Fernet <2 ms; JWT ~0.5 ms

Full raw run IDs and methodology are in the Validation & Timing Report; every figure is reproducible with two commands from the repository.

5. Why Sentinel Stands Out

- Ecosystem, not app: three synchronized interfaces over one state layer.
- AI that assists but never decides - explainable, deterministic safety paths.
- On-premises + offline inference = data never leaves the clinic.
- Works on consumer smart rings and runs at zero monthly operating cost.

6. Stage and Ask

Sentinel is an independent research project with a working, fully tested implementation. The current ask of an incubation program is: infrastructure support for live hardware pilots, clinical-partner access for validation studies, and mentorship on regulatory pathways (clinical software, privacy compliance). Milestones and the commercial plan are detailed in the Business Plan (06).